

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I _SUDHARSANAN_NARASHIMAN 192524061 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and**

trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT

enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

and
coherence

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1. File Transfer with Transport and Data Link Layers

Given:

- File size = **150 MB = 150,000,000 bytes**
- Segment size = **1500 bytes**
- Data Link overhead/frame = **40 bytes**
- Bandwidth = **50 Mbps**

Step 1: Number of segments

$$150,000,000 / 1500 = 100,000 \quad \frac{150,000,000}{1500} = 100,000$$

Segments = 100,000

Since each segment becomes one frame,

Frames = 100,000

Step 2: Total overhead

$$100,000 \times 40 = 4,000,000 \text{ bytes} \quad 100,000 \times 40 = 4,000,000 \text{ bytes}$$

Overhead = 4 MB

Step 3: Total transmitted data

$$150 + 4 = 154 \text{ MB} \quad 150 + 4 = 154 \text{ MB}$$

Step 4: Transmission time

154 MB

Convert to bits

$$154 \times 8 = 1232 \text{ Mb} \quad 1232 / 50 = 24.64 \text{ s} \quad \frac{1232}{50} = 24.64 \text{ s}$$

Answer

- Segments = **100,000**
- Frames = **100,000**
- Overhead = **4 MB**
- Transmission time = **24.64 s**

Explanation

- **Transport Layer:** Segments the file and provides reliable delivery.
- **Data Link Layer:** Frames the segments, performs error detection, and enables reliable local delivery.

2. Sliding Window

Given

- Window size = **6**
- Total frames = **48**

Windows required

$$48 \div 6 = \frac{48}{6} = 8$$

Retransmissions

One frame lost per window.

$$8 \times 1 = 8$$

Answer

- Windows = **8**
- Retransmissions = **8**

Explanation

Flow control prevents sender overflow by matching transmission speed with receiver capacity, improving efficiency.

3. IP Fragmentation

Given

- Packet = **12,000 bytes**
- MTU = **1500 bytes**
- IP Header = **20 bytes**

Maximum payload per fragment

$$1500 - 20 = 1480 \text{ bytes}$$

Fragments

$$\lceil \frac{12000}{1480} \rceil = 9 \left\lceil \frac{12000}{1480} \right\rceil = 9$$

Header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Answer

- Fragments = **9**
- Header overhead = **180 bytes**

Explanation

The Network Layer fragments packets when necessary and reassembles them at the destination using fragment information.

4. Sliding Window

Answer

- Windows = **8**
- Retransmissions = **8**

Explanation

Flow control ensures smooth communication by preventing congestion and receiver overflow.

5. Session Layer Synchronization

Given

- File = **600 MB**
- Checkpoint every **60 MB**
- Failure after **390 MB**

Checkpoints

60,120,180,240,300,360

Total

6 checkpoints

Data retransmitted

$$390 - 360 = 30 \text{ MB}$$

Answer

- Checkpoints = **6**
- Retransmitted data = **30 MB**

Explanation

Synchronization checkpoints allow communication to resume from the latest checkpoint instead of restarting.

6. Presentation Layer

Given

Original file = **900 MB**

Compression = **40%**

Compressed size

$$900 \times 0.60 = 540 \text{ MB}$$

Encryption overhead

$$540 \times 1.05 = 567 \text{ MB}$$

Reduction

$$\frac{900 - 567}{900} \times 100 = 37\%$$

Answer

- Compressed size = **540 MB**
- Encrypted size = **567 MB**
- Total reduction = **37%**

Explanation

The Presentation Layer compresses data to reduce transmission time and encrypts it to ensure

confidentiality.

7. FTP Application

Given

- Data = **3 GB = 3000 MB**
- Request size = **3 MB**
- Throughput = **120 Mbps**

Requests

$$\frac{3000}{3} = 1000 \quad \frac{3000}{3} = 1000$$

Transmission time

3000 MB

$$\frac{3000 \times 8}{120} = 200 \quad \frac{3000 \times 8}{120} = 200$$

Answer

- Requests = **1000**
- Transmission time = **200 s**

Explanation

The Application Layer provides user services such as FTP, file management, and reliable communication.

8. End-to-End Delay

Processing delay

Application

4 ms

Transport

3 ms

Network

5 ms

Data Link

2 ms

Physical

1 ms

Total processing

$4+3+5+2+1=15\text{ ms}$

Propagation delay

18 ms

Transmission delay

12 ms

Total delay

$15+18+12=45\text{ ms}$

Answer

Total end-to-end delay = 45 ms

Explanation

Each OSI layer processes the packet before it is transmitted across the network.

9. Protocol Overhead

Given

Useful data

80 MB

Frame size

1600 bytes

Overhead

80 bytes

Useful payload

$$1600 - 80 = 1520 \text{ bytes}$$

Frames

$$\lceil \frac{80,000,000}{1520} \rceil = 52,632$$

Overhead

$$52,632 \times 80 = 4,210,560 \text{ bytes}$$

$\approx 4.21 \text{ MB}$

Efficiency

$$\frac{1520}{1600} \times 100 = 95\%$$

Answer

- Frames = **52,632**
- Overhead = **4,210,560 bytes ($\approx 4.21 \text{ MB}$)**
- Efficiency = **95%**

Explanation

Protocol overhead supports addressing, error checking, and control, but reduces transmission efficiency.

10. Hospital IoT Network

Given

Original file

240 MB

Compression

30%

Compressed size

$$240 \times 0.70 = 168 \text{ MB}$$

Compressed bytes

168,000,000 168,000,000 168,000,000

Segments

$168,000,000 / 1400 = 120,000$ $\frac{168,000,000}{1400} = 120,000$ $168,000,000 / 1400 = 120,000$

Each frame overhead

60 bytes

Protocol overhead

$120,000 \times 60 = 7,200,000$ bytes $120,000 \times 60 = 7,200,000$ bytes $120,000 \times 60 = 7,200,000$ bytes

= 7.2 MB

Final transmitted data

$168 + 7.2 = 175.2$ MB $168 + 7.2 = 175.2$ MB $168 + 7.2 = 175.2$ MB

Answer

- Compressed file size = **168 MB**
- Segments = **120,000**
- Protocol overhead = **7.2 MB**
- Total transmitted data = **175.2 MB**

Explanation

- **Presentation Layer:** Compresses data and can provide encryption.
- **Transport Layer:** Segments data for reliable delivery.
- **Data Link Layer:** Adds frame headers/trailers and detects errors.
- **Physical Layer:** Transmits the final data as bits through the network.

These layers work together to provide **efficient, secure, and reliable communication**.